MIL-R-13442C
1 August 1967

SUPERSEDING
MIL-R-13442B
4 September 1962

MILITARY SPECIFICATION

REELING MACHINE, CABLE, HAND-

ADJUSTABLE AND COLLAPSIBLE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers and adjustable, collapsible cable reel operated manually by a hand crank.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Box, Fiberboard.

Military

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U. S. Military Property.

FSC 3895

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained form the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN TRUCKING ASSOCIATIONS, INC.

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street NW, Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Description. The reeling machine shall consist of a stand, a reel, and a hand crank. The stand shall be collapsible and the reel and crank shall be removable. The reeling machine shall have the dimensions and configuration as shown in figure 1 and shall operate as specified herein.

3.2 Preproduction model. The supplier shall furnish a reeling machine within the time frame specified in the contract or purchase order to prove, prior to starting production, that his production methods and choice of design detail will produce reeling machines that comply with the requirements of this specification. Examination and tests shall be those specified herein. Any changes or deviations from the preproduction model shall be subject to the approval of the contracting officer. Approval of the preproduction model by the activity concerned shall not relieve the supplier of his obligation to furnish reeling machines conforming to this specification.

3.3 Initial production. When specified (see 6.2), the supplier shall furnish to the Government one or more reeling machines (see 6.2) for testing as specified in 4.4.

3.4 Material. Material shall be as specified herein. Material not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

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3.5 Stand. The stand shall be constructed of 1/8-inch-thick AISI 1020 steel and shall be collapsible as shown in figure 2. The stand shall rest on not less than eight legs and shall support the reel and hand crank when set up in operating position. Provisions shall be incorporated so that the reel and hand crank may be removed from the stand, the hand crank may be secured to the stand when the stand is in the collapsed position, and the reel and hand crank may be retained in the stand during winding and unwinding operations. The stand shall incorporate bearings to provide wearing surfaces at points of rotation. The stand shall be gusseted or otherwise reinforced where necessary to provide strength. Dimensions and configuration shall be as shown in figure 1.

3.6 Reel. The reel shall consist of drive and driven flanges and reel arms. Eight reel arms shall be provided and shall be riveted to the flanges. The flanges and reel arms shall be adjustable as shown in figure 1. Provision shall be made to hold the reel and reel arms in any position from the fully extended position to the fully retracted position. The configuration and required dimensions of the flanges shall be as shown in figure 3. The dimensions of the reel arms shall be as shown in figure 4.

3.7 Crank. A hand crank shall be provided and shall drive the reel and reel arms during winding and unwinding operations.

3.8 Performance. The reeling machine shall permit the winding and unwinding of a 100-foot length of No. 4 AWG or No. 8 AWG wire onto and off the reel without chattering or swaying during the winding or unwinding operations. The reel arms shall not bind, chafe, or distort during the winding or unwinding operations.

3.9 Identification marking. The reeling machine shall be identified and permanently marked in accordance with MIL-STD-130.

3.10 Treatment and painting. Unless otherwise specified (see 6.2), the portions of the reeling machine normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A.

3.11 Workmanship. All parts, components, and assemblies of the reeling machine shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. All exposed edges shall be rounded or chamfered.

3.11.1 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size

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for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.11.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided, and all bolts, nuts, and screws shall be tight.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of preparation for delivery (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The reeling machine shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The reeling machine shall be tested as specified in 4.6.2. Failure of the test shall be cause for rejection.

4.4 Initial production inspection. An initial production reeling machine(s) will be selected at random from the first ten reeling machines being produced from production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production reeling machine shall not exclude the remaining reeling machines from the quality conformance inspection and acceptance provisions specified in section 4.

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4.4.1 Inspection failure. Failure of an initial production reeling machine to meet any requirement specified herein during and as a result of the examination and test specified in 4.4 shall be cause for rejection of the initial production reeling machine(s) and shall be cause for refusal by the Government to continue acceptance of production reeling machines until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on reeling machines previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all reeling machines accepted prior to the completion of the initial production inspection are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each reeling machine shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.6 Inspection procedure.

4.6.1 Examination. The reeling machine shall be examined for the following defects:

101. Material not as specified.
102. Dimensions not as specified.
103. Stand, reel, or hand crank not as specified.
104. Identification marking not as specified.
105. Treatment and painting not as specified.
106. Workmanship not as specified.

4.6.2 Test. Erect the reeling machine, wind and unwind a 100-foot length of No. 4 AWG wire onto and off the reel, and wind and unwind a 100-foot length of No. 8 AWG wire onto and off the reel. Determine whether the reeling machine sways or chatters during the winding and unwinding operations. At the end of the unwinding phases, examine the reel arms for damage. Inability of the reeling machine to operate; chattering or swaying of the reeling machine during operation; or binding, chafing, distortion or other damage to the reel arms shall constitute failure of this test.

4.7 Inspection comparison. The Government may select reeling machines at any time during the contract production period and subject these reeling machines to the examination specified in 4.6.1 and to the test specified in 4.6.2 to determine that the quality of the selected reeling machine is equal to the quality standards established during initial production inspection. The inspection will be performed by the Government at a site selected by the Government. Reeling machines will be selected at random

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from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production reeling machines.

4.7.1 Inspection failure. Failure of any reeling machine when inspected as specified in 4.7 may be considered cause for refusal by the Government to continue acceptance of reeling machines until objective evidence furnished by the supplier reveals that corrective action has been taken to eliminate the condition which caused the rejection.

4.8 Inspection of preparation for delivery.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For the purpose of inspection, a completely processed pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Samples selected in accordance with 4.8.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107.	Materials, methods, or containers not as specified for level A or B. Each incorrect material, methods, or container shall be considered one defect.
108.	Reeling machine not collapsed as specified for level A or B.
109.	Unpainted surfaces not coated as specified for level A or B.
110.	Packing not as specified for level A or B.
111.	Strapping not zinc coated for level A.
112.	Marking illegible, incomplete, or incorrect.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each reeling machine shall be collapsed as shown in figure 2. Unpainted metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative conforming to the applicable specification listed in and applied in accordance with MIL-P-116.

5.1.2 Level C. Each reeling machine shall be collapsed as shown in figure 2. The collapsed machine shall be preserved and packaged in a manner to afford protection against deterioration and damage from the supplier to the initial destination.

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5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Each reeling machine with its components and parts shall be packed in a box conforming to PPP-B-601, overseas type, style optional. Blocking, bracing, and cushioning shall be provided to prevent movement of the contents. The boxes shall be closed and strapped in accordance with the appendix to the box specification. Strapping shall be zinc coated.

5.2.2 Level B. Each reeling machine with its components and parts shall be packed in a box conforming to PPP-B-636, V3c, style optional. The box shall be waterproof sealed with tape and reinforced with tape in accordance with the appendix to the box specification.

5.2.3 Level C. The reeling machine with its components and parts shall be packed to insure carrier acceptance and safe delivery at lowest rates in containers complying with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.3 Marking. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The reeling machine is intended for use in holding and dispensing electric lighting equipment wire.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When initial production testing is required (see 3.3).
(c) Number of reeling machines to be furnished for initial production testing, when applicable (see 3.3).
(d) When treatment and painting shall be other than as specified (see 3.10).
(e) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

6.3 Revision changes. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - ME
Air Force - 84

Review activities:

Army - EL
Air Force - 84

User activities:

Army - CE

Preparing activity:

Army - ME

Code C

Project No. 3895-0156

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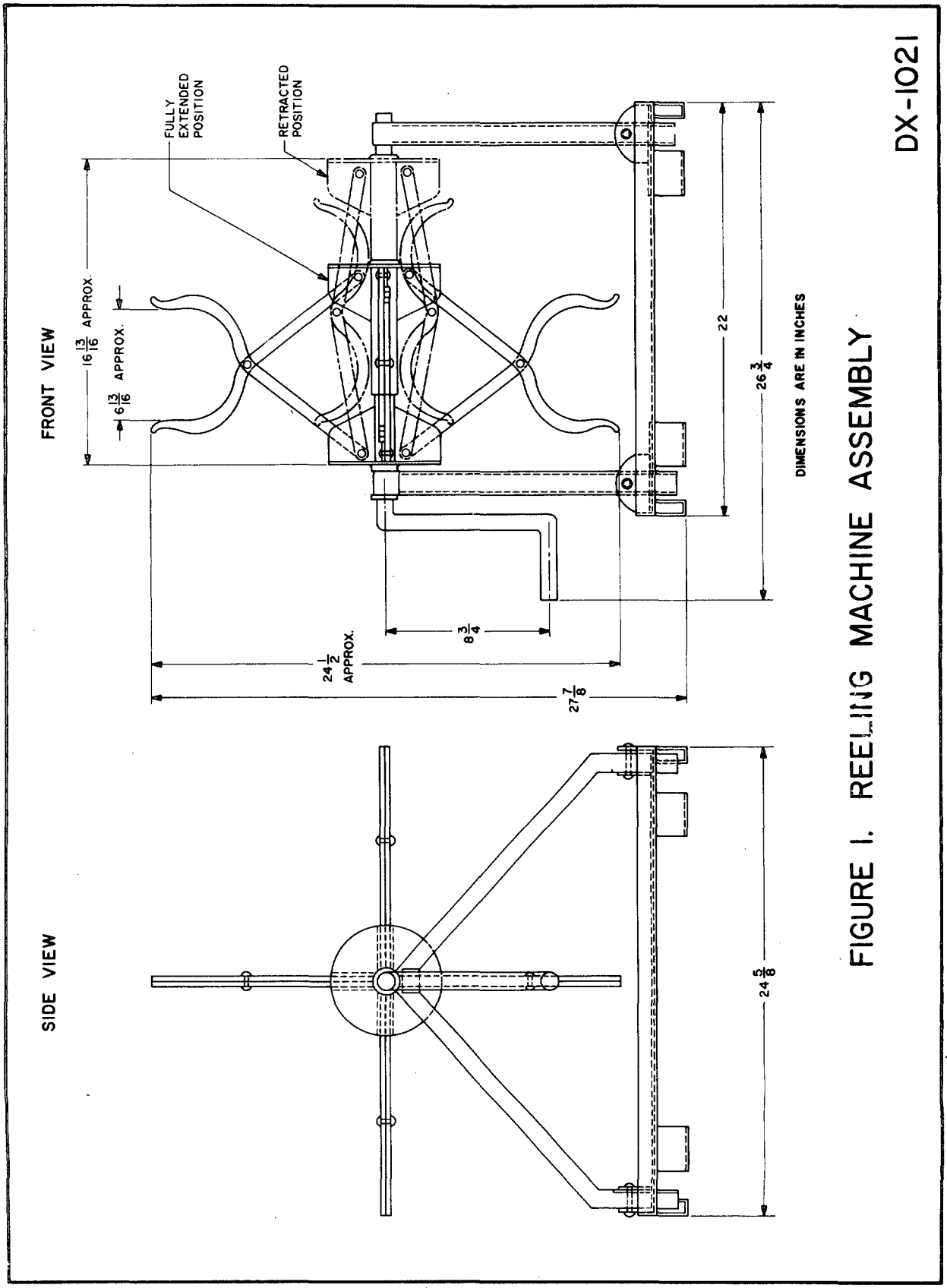


FIGURE I. REELING MACHINE ASSEMBLY

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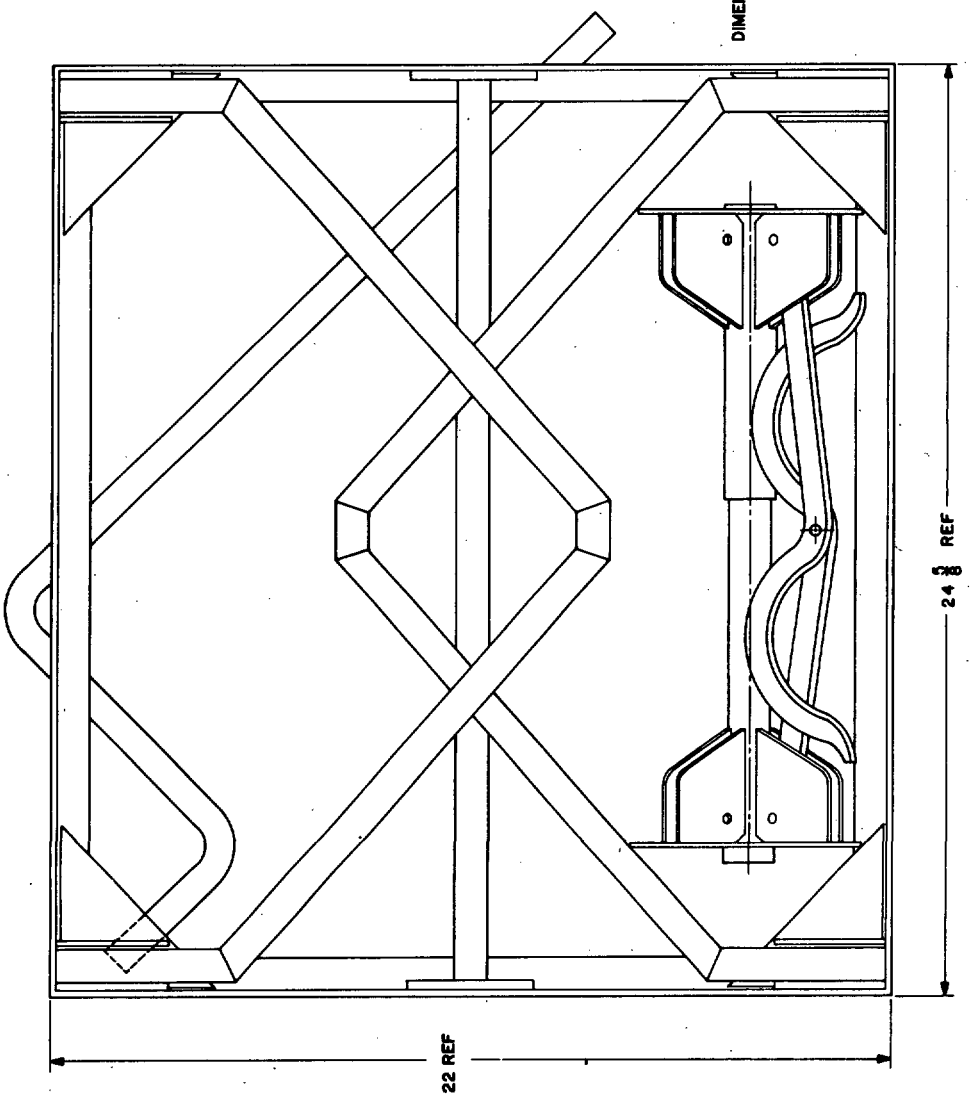
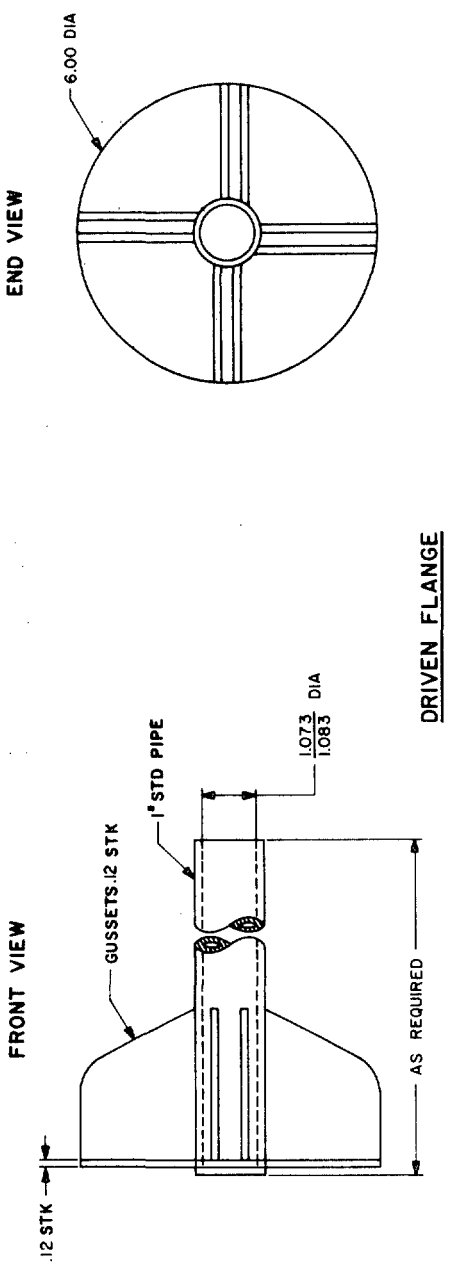
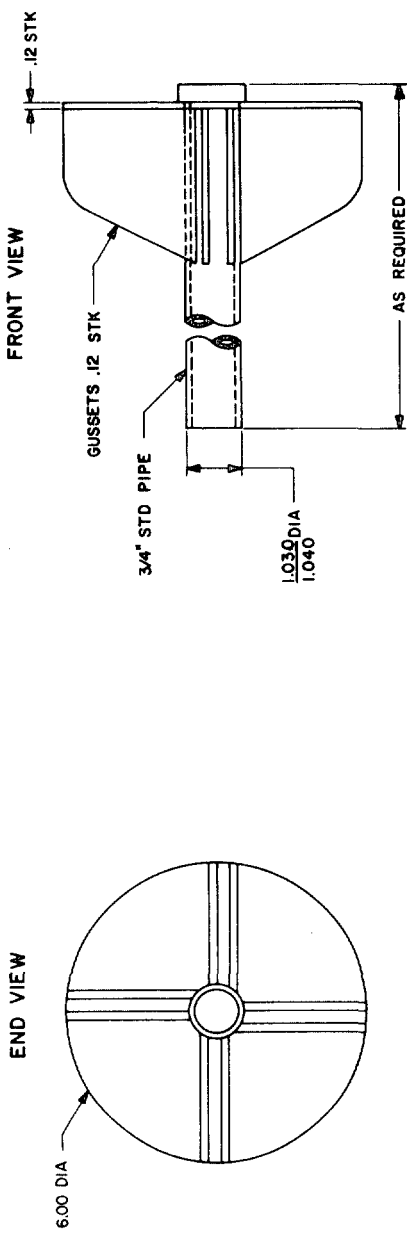


FIGURE 2. REELING MACHINE IN COLLAPSED POSITION

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DX-1023

FIGURE 3. FLANGES

NOTES:
1. DIMENSIONS ARE IN INCHES
2. MATERIAL: STEEL, AISI 1020.

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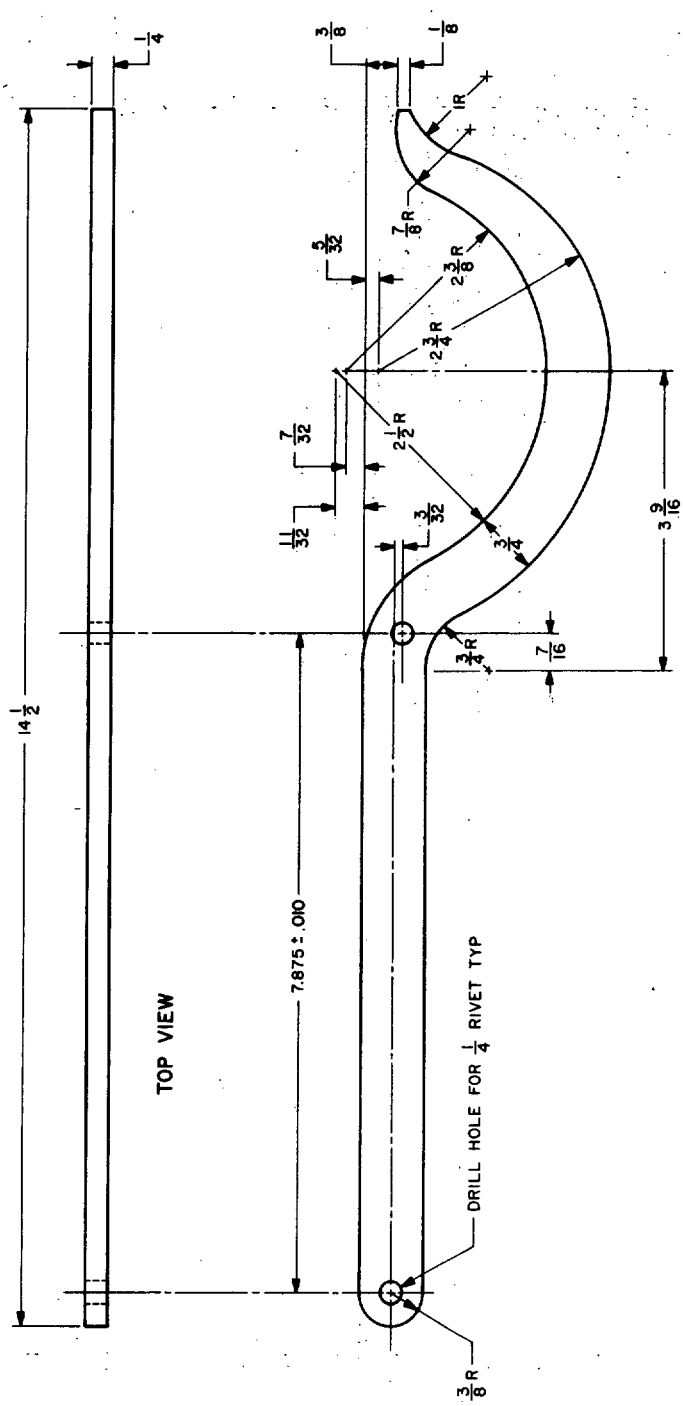


FIGURE 4. REEL ARM

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U. A. ARMY MOBILITY EQUIPMENT COMMAND
RESEARCH, DEVELOPMENT AND ENGINEERING DIRECTORATE
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FORT BELVOIR, VIRGINIA 22060

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-R-13442C - Reeling Machine, Cable, Hand-Adjustable & Collapsible

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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